

Praveena Kurukuladithya

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OBJECTIVE

Computer Systems Engineering undergraduate with hands-on experience in **embedded systems, Industrial IoT, robotics, PLC automation, PCB design, and Linux-based development**. Experienced in building hardware-software systems using **ESP32, Arduino, Raspberry Pi, sensors, motor drivers, MQTT, Flask/FastAPI dashboards, and control algorithms**. Seeking an **Embedded Systems, IoT, Robotics, or Industrial Automation internship** to apply practical engineering skills in real-world product development, testing, troubleshooting, and deployment.

TECHNICAL SKILLS

Embedded & IoT: Arduino, ESP32, ESP32-S3, Raspberry Pi, GPIO, UART, I2C, SPI, PWM, ADC, Timers, Interrupts, MQTT, WebSocket, Sensor Interfacing

Robotics & Control: DC Motor Control, Encoder Feedback, PID Control, Line Following, Obstacle Detection, Motor Drivers, TB6612FNG, L298N, Real-Time Control Logic

PCB & Electronics: EasyEDA, Proteus, PCB Design, PCB Etching, Dot Board Soldering, Soldering, Circuit Design, Debugging, Multimeter Testing, Power Distribution

Languages: C, C++, Python, JavaScript, HTML/CSS, Bash, AVR Assembly, SQL

Web & Software: Flask, FastAPI, MERN Stack Basics, Node.js, Express.js, React, Firebase, REST APIs, Chart.js, Git, GitHub

Industrial Automation: Siemens S7-200, Siemens S7-1200, PLC Programming, PLC Wiring, Ladder Logic Basics, Automation Troubleshooting

Linux & DevOps: Raspberry Pi OS, Linux Terminal, SSH, Docker, Basic CI/CD, Web App Deployment

PROJECTS

IIoT Predictive Maintenance System — ESP32, Raspberry Pi, AI/ML, Docker, IoT Dashboard

- Designed an **Industrial IoT predictive maintenance architecture** using ESP32 as the embedded controller layer and Raspberry Pi as the edge computing and dashboard layer.
- Integrated machine-condition monitoring concepts using **vibration, temperature, and current sensor data** to support early fault detection and maintenance decisions.
- Planned an **AI/ML-based fault prediction workflow** for identifying abnormal machine behavior from collected sensor patterns.
- Developed the system concept with **IoT dashboard control, actuator/motor control, Linux-based Raspberry Pi operation, Docker deployment, MQTT communication, and PCB-level integration**.

PID Motor Control System & IIoT SCADA — Raspberry Pi, Python, Flask, Encoder

- Built a **closed-loop DC motor speed control system** using Raspberry Pi, 6V N20 DC motor, 700-PPR optical encoder, and TB6612FNG motor driver.
- Derived an empirical motor transfer function through **open-loop step response testing, system identification, and PID performance analysis**.
- Developed a **Flask-based SCADA dashboard** with Chart.js for live RPM monitoring, PID tuning, setpoint control, and emergency stop operation.
- Implemented real-time **PID control in Python** with anti-windup protection, dynamic delta-time calculation, and PWM control to reduce Linux timing-jitter effects.
- Applied **DSP moving-average filtering** to reduce encoder quantization noise and automated data logging/graphing using pandas, matplotlib, and scipy.

Autonomous Line Following, Obstacle Detection & Parking Robot — Arduino, AVR Assembly

- Developed an Arduino Uno-based autonomous robot car using **AVR Assembly** for low-level sensor reading, decision-making, and motor control.
- Interfaced **IR sensors** for line detection, an ultrasonic sensor for obstacle detection, and TB6612FNG motor driver for differential movement.
- Implemented **priority-based control logic** where obstacle detection overrides movement, allowing the robot to stop safely and resume when the path is clear.

- Designed state-machine-based logic for **line following, left/right correction, parking-zone detection, and automatic parking sequence execution**.
- Used Timer0-based **PWM motor speed control** and ATmega328P digital I/O pins for direction control and real-time movement handling.

BJT Audio Amplifier Circuit — Analog Electronics, PCB, Signal Amplification

- Designed and built a **common-emitter BJT audio amplifier** for analog voice/audio signal amplification with approximately **20x voltage gain**.
- Selected and tested biasing resistors, coupling capacitors, and emitter bypass components to achieve stable amplification and suitable operating point.
- Practiced **circuit design, soldering, waveform observation, debugging, and analog signal behavior analysis** using practical lab testing methods.

EDUCATION

Sri Lanka Institute of Information Technology (SLIIT)

Nov. 2023 – Present

BSc (Hons) in Computer Systems Engineering

Malabe, Sri Lanka

- **Relevant Areas:** Embedded Systems, Control Systems, Digital Systems, DSP, Electronics, IoT, Robotics, Programming

Kalutara Vidyalyaya

2008 – 2022

GCE Advanced Level — Physical Science Stream | Z-Score: 0.6896

Kalutara, Sri Lanka

CERTIFICATIONS & QUALIFICATIONS

Designing, Maintaining & Troubleshooting Automated Engineering Systems — PLC

Nov. 2025 – Feb. 2026

Robotics & Control Systems Lab, Dept. of Mechanical Eng., Faculty of Engineering

University of Moratuwa, Sri Lanka

- Completed hands-on training in **PLC programming, PLC wiring, industrial automation troubleshooting, and automated system maintenance**.
- Practiced **Siemens S7-200 and S7-1200 PLC programming** with real industrial automation wiring and control scenarios.

LEADERSHIP & ACTIVITIES

Batch Representative — Computer Systems Engineering Batch 02

Jan. 2024 – Present

Department of Computer Systems Engineering, SLIIT

Malabe, Sri Lanka

- Represent the batch and coordinate academic communication, student concerns, department updates, and event-related announcements.

Executive Committee Member

Aug. 2025 – Present

Computer Systems Engineering Student Committee, SLIIT

Malabe, Sri Lanka

- Support planning and coordination of department-level student activities, technical events, and community engagement initiatives.

Event Coordinator / Committee Member

2024 – Present

IdeaSpark 2 & 3, AGMs, CodeFest 2025, Innova IoT

Sri Lanka

- Contributed to event planning, communication, participant coordination, and technical/community engagement activities.

SOFT SKILLS

Problem Solving | Practical Debugging | Teamwork | Leadership | Communication | Time Management | Self-Learning | Adaptability | Technical Documentation | Project Coordination

REFERENCES

Prof. Anuradha Jayakody

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